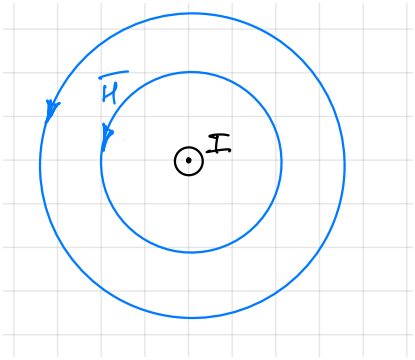
ELECTROMAGNETISM: ANSWERS

This is a closed-book examination. A formula sheet is provided

Question 1. [a) = 5] [b) = 10] [c) = 10] [d) = 10] [e) = 5]

a) (bookwork and reasoning)

The field lines are concentric circles around the current [2]. The direction of the current determines the direction of the field. From Maxwell's equation for static magnetic fields $\nabla \times \mathbf{H} = I$, we see that reversing the direction of the current ($I \to -I$) reverses the direction of the field, $\mathbf{H} \to -\mathbf{H}$ [3].



b) (bookwork) Assuming static fields, $\partial/\partial t = 0$, we see that four Maxwell's equations are split into two independent pairs, for electric and for magnetic quantities [2]. The relevant equation for static electric fields is $\nabla \times \mathbf{E} = 0$ [1]. It follows that field $\mathbf{E}$ is conservative and can be represented as $\mathbf{E} = -\nabla\varphi$, where φ is the electrostatic potential [2]. Voltage, or potential difference, between two points, A and B, is then introduced as $V = \varphi(A) - \varphi(B)$ [1]. On the other hand, integrating

$$\int_A^B \mathbf{E} \cdot d\mathbf{l} = -\int_A^B (\nabla\varphi) \cdot d\mathbf{l} = \varphi(A) - \varphi(B) = V \qquad [3]$$

The value of the integral thus does not depend on the path chosen between points A and B [1].

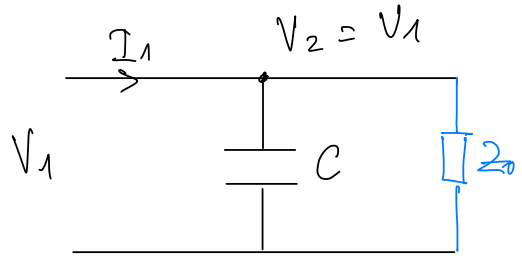
c) (bookwork) For time varying fields, $\nabla \times \mathbf{E} \neq 0$ [2]. Instead, $\nabla \times \mathbf{E} = -\partial\mathbf{B}/\partial t$ [2]. Therefore, field $\mathbf{E}$ is not conservative in general and *cannot* be represented as a gradient of a potential [2]. The value of the integral $\int_A^B \mathbf{E} \cdot d\mathbf{l}$ will, in general, depend on the path chosen [2]. It implies that voltage between two points may not be a practically useful concept [2].

d) (calculated example)

Refer to the figure below for notation.

The device is symmetric, and therefore $S_{11} = S_{22}$ and $S_{21} = S_{12}$ [1]. We use the following equations (from the formula sheet provided)

$$S_{11} = \frac{V_1 - Z_0 I_1}{V_1 + Z_0 I_1} \qquad S_{21} = \frac{2V_2}{V_1 + Z_0 I_1}$$



Using standard analysis, $I_1 = V_1/Z$ where Z is a parallel combination of the capacitor and Z_0:

$$Z = \frac{Z_0}{1 + j\omega Z_0 C} \quad [1]$$

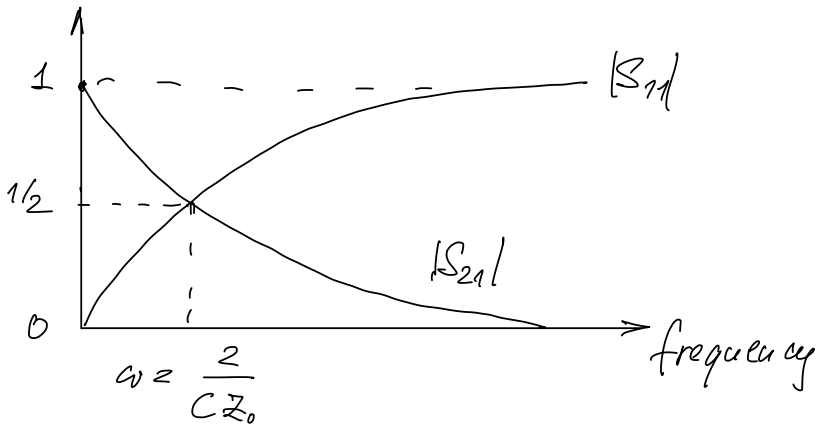
Also, $V_2 = V_1$ [1]. Then

$$S_{11} = -\frac{j\omega Z_0 C}{2 + j\omega Z_0 C} \qquad S_{21} = \frac{2}{2 + j\omega Z_0 C} \qquad [3]$$

We see that $|S_{11}|^2 + |S_{21}|^2 = 1$, so that power is conserved (as it should in a passive device) [1].

We see that $|S_{11}| \approx 0$ and $|S_{21}| \approx 1$ at low frequencies; and $|S_{11}| \approx 1$ and $|S_{21}| \approx 0$ at high frequencies. Also, $|S_{11}| = |S_{21}| = 1/2$ when $\omega = 2/(Z_0 C)$ [1].

The graphs plotted below show the device is a low-pass filter. [2]



e) (bookwork) In far-field, the antenna has two field components, E_θ and H_φ. Expressions are given in the formula sheet. Power flow is by definition

$$\mathbf{P} = \frac{1}{2}\mathrm{Re}[\mathbf{E} \times \mathbf{H}^*] \quad [2]$$

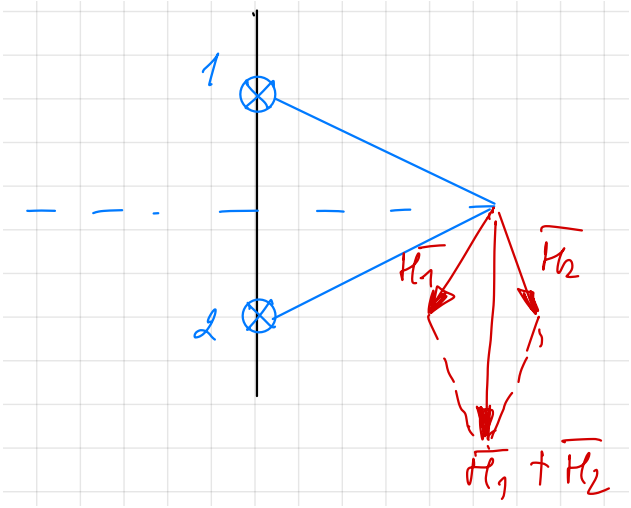
The direction will be given by $\mathbf{e}_\theta \times \mathbf{e}_\varphi = \mathbf{e}_r$ [3].

Question 2. [ai) = 5], [aii) = 10], [bi) = 5], [bii) = 10]

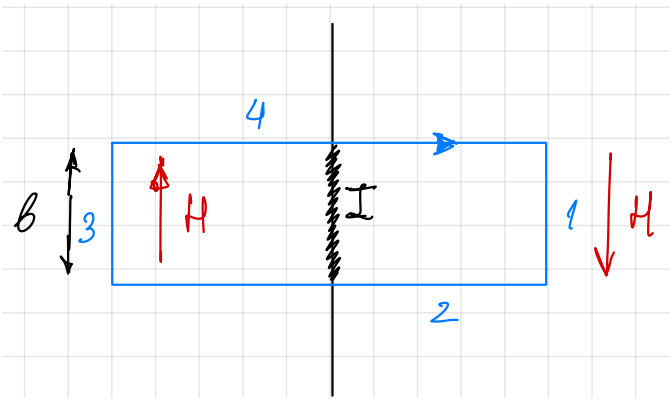
All questions are reasoning/calculated examples.

a)

i) We split the plane into thin strips, each of which can be seen as an infinitely long current [1]. Choosing two strips and looking at the symmetry line as shown, we see that the field lines of the pair are vertical [2]. The same applies to any pair of strips, so that the entire field lines will be vertical [1]. Their direction is opposite on the opposite sides of the plane [1].



ii) The shortest derivation is by Ampere's law in the integral form. Choose a rectangular path as shown [3].



We get

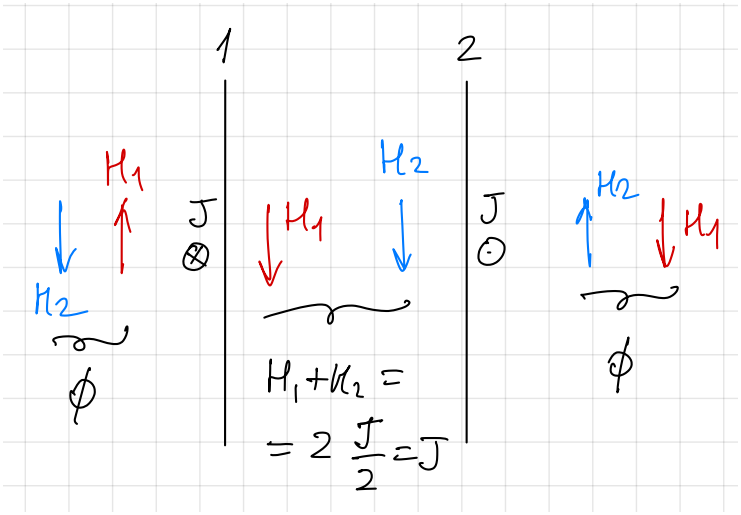
$$\oint \mathbf{H}\cdot d\mathbf{l} = \int_1 \mathbf{H}\cdot d\mathbf{l} + \int_2 \mathbf{H}\cdot d\mathbf{l} + \int_3 \mathbf{H}\cdot d\mathbf{l} + \int_4 \mathbf{H}\cdot d\mathbf{l} = Hb + 0 + Hb + 0 \qquad [4]$$

On the other hand, this integral equals the total current enclosed, namely $I = Jb$ [2].

Hence $H = J/2$ [1].

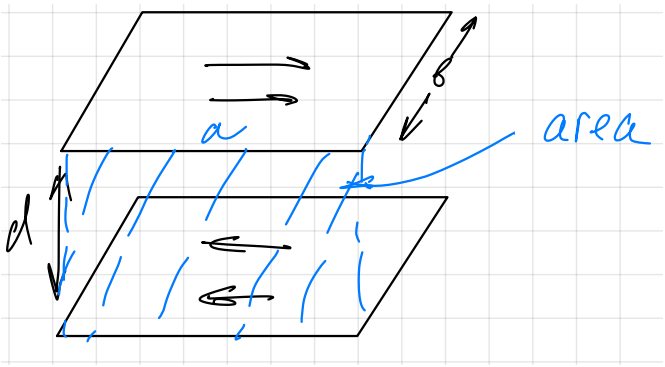
b)

i) We rely on the result of a)ii) and notice that the field reverses direction when the current reverses direction [1]. Therefore, the field is zero to the left and to the right of the planes [2]. Between the planes it is $H = 2J/2 = J$ [2].



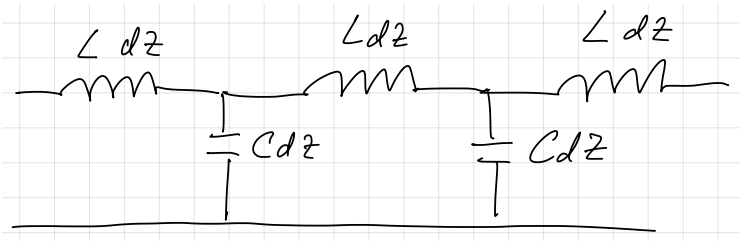
ii) Choosing an area as shown [3], we obtain $\Phi = \iint \mathbf{B} \cdot d\mathbf{S} = \mu_0 J d a$ and $I = bJ$ [4]. Then per-unit-length inductance is

$$L = \frac{\Phi}{aI} = \mu_0 \frac{d}{b} \qquad [3]$$



Question 3. [a) = 10], [b) = 10], [c) = 10]

a) Given the capacitance and inductance are distributed, an equivalent circuit can have the form shown [2]. This is the circuit of a transmission line [1].

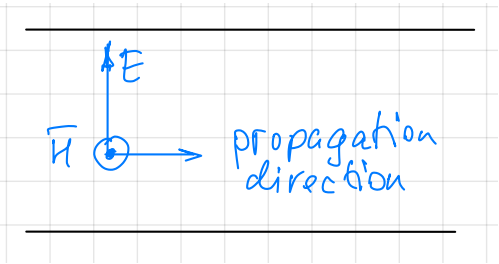


The wave velocity is $c = \frac{1}{\sqrt{LC}}$ [3]. Therefore

$$\frac{1}{\sqrt{L\varepsilon_0 b/d}} = \frac{1}{\sqrt{\varepsilon_0\mu_0}} \qquad [2]$$

and $L = \frac{\mu_0 d}{b}$ [2].

b) Consider a TEM wave that propagates along the waveguide. This wave has two field components, both of which are perpendicular to the propagation direction [2]. We choose the direction of the electric field to be perpendicular to the plates [2]. Then the boundary conditions on the PECs are satisfied automatically [2]. It means that such a wave can indeed propagate along the waveguide [1]. It has the same linear dispersion $\omega = ck$ and the same wave velocity c as the wave propagating along the transmission line [3].



c) The TEM wave has a linear dispersion curve. The TM and TE waves have parabolic-like dispersion curves [1] that are identical to each other [2], as shown in the figure below [2]. The frequency cut-off appears because both TM and TE modes have an electric field component along the conducting plates [2]. At the plates the boundary conditions require that this field component vanishes [1], leading to standing waves in the transverse direction that have a discrete frequency spectrum [2].

